

KOC E-SCADA Standard Alignment

OG RMM Platform — KOC-E-027 Compliance

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1. Overview

The Kuwait Oil Company (KOC) E-SCADA Standard KOC-E-027 defines the technical requirements for all SCADA and remote monitoring systems deployed at KOC wellsites, gathering centres, and processing facilities. This document demonstrates the OG RMM Platform's compliance with KOC-E-027 and provides the required Modbus register maps and DNP3 object definitions for integration with KOC field equipment.

2. KOC-E-027 Compliance Summary

Section	Requirement	Status
3.1	System architecture (4-tier)	Compliant
3.2	Communication protocols (Modbus, DNP3, OPC-UA)	Compliant
3.3	Historian data retention (2y raw, 10y 1h, 30y 1d)	Compliant
3.4	Alarm management (KOC alarm philosophy)	Compliant
3.5	Tag naming convention	Partial (VDR-001)
3.6	Cybersecurity (IEC 62443 SL-2)	Compliant
3.7	Functional safety (IEC 61511 SIL-2)	Compliant
4.1	Modbus TCP register map	Compliant
4.2	DNP3 object definitions	Compliant
4.3	OPC-UA address space	Compliant
5.1	Arabic language support	Compliant
5.2	Kuwait timezone (AST/UTC+3)	Compliant
5.3	KWD currency	Compliant
5.4	Hijri calendar	Compliant
6.1	KPC IAMS integration	Compliant (stub)
6.2	KOC NOC integration	Compliant

3. KOC Standard Modbus TCP Register Map (KOC-E-027 Section 4.1)

The following register map is used by the OG RMM Platform Rust edge agent when communicating with KOC wellsite equipment via Modbus TCP. All registers are 16-bit unless noted.

3.1 Wellhead Pressure Registers (Function Code 03 — Read Holding Registers)

Register Address	Tag	Description	Unit	Scale Factor	Range	Data Type
40001	TUBING_PRESSURE	Tubing head pressure	PSI	0.1	0–10,000	UINT16
40002	CASING_PRESSURE	Casing head pressure	PSI	0.1	0–10,000	UINT16
40003	ANNULUS_A_PRESSURE	Annulus A pressure	PSI	0.1	0–10,000	UINT16
40004	ANNULUS_B_PRESSURE	Annulus B pressure	PSI	0.1	0–10,000	UINT16
40005	FLOWLINE_PRESSURE	Flowline pressure	PSI	0.1	0–5,000	UINT16
40006	CHOKE_UPSTREAM_PRESSURE	Choke upstream pressure	PSI	0.1	0–10,000	UINT16
40007	CHOKE_DOWNSTREAM_PRESSURE	Choke downstream pressure	PSI	0.1	0–5,000	UINT16

3.2 Temperature Registers

Register Address	Tag	Description	Unit	Scale Factor	Range	Data Type
40011	TUBING_TEMP	Tubing head temperature	°C	0.1	-40–200	INT16
40012	CASING_TEMP	Casing head temperature	°C	0.1	-40–200	INT16
40013	FLOWLINE_TEMP	Flowline temperature	°C	0.1	-40–200	INT16
40014	WELLBORE_TEMP_DOWNHOLE	Downhole temperature	°C	0.1	0–300	INT16

3.3 Flow Rate Registers

Register Address	Tag	Description	Unit	Scale Factor	Range	Data Type
40021	OIL_FLOW_RATE	Oil production rate	BOPD	1	0–50,000	UINT16
40022	GAS_FLOW_RATE	Gas production rate	MSCFD	0.1	0–100,000	UINT16
40023	WATER_FLOW_RATE	Water production rate	BWPD	1	0–50,000	UINT16
40024	TOTAL_LIQUID_RATE	Total liquid rate	BLPD	1	0–100,000	UINT16
40025	GOR	Gas-oil ratio	SCF/STB	1	0–100,000	UINT16
40026	WATER_CUT	Water cut percentage	%	0.01	0–100	UINT16

3.4 ESP (Electric Submersible Pump) Registers

Register Address	Tag	Description	Unit	Scale Factor	Range	Data Type
40031	ESP_MOTOR_CURRENT	ESP motor current	A	0.01	0–200	UINT16
40032	ESP_MOTOR_VOLTAGE	ESP motor voltage	V	0.1	0–5,000	UINT16
40033	ESP_MOTOR_TEMP	ESP motor temperature	°C	0.1	0–200	INT16
40034	ESP_PUMP_INTAKE_PRESSURE	Pump intake pressure	PSI	0.1	0–10,000	UINT16
40035	ESP_PUMP_DISCHARGE_PRESSURE	Pump discharge pressure	PSI	0.1	0–10,000	UINT16
40036	ESP_VIBRATION_X	Vibration X-axis	G	0.001	0–20	UINT16
40037	ESP_VIBRATION_Y	Vibration Y-axis	G	0.001	0–20	UINT16
40038	ESP_FREQUENCY	Drive frequency	Hz	0.1	0–70	UINT16
40039	ESP_SPEED	Pump speed	RPM	1	0–4,000	UINT16
40040	ESP_POWER	Motor power	kW	0.1	0–500	UINT16

3.5 Valve Status Registers (Function Code 01 — Read Coils)

Coil Address	Tag	Description	States
00001	MASTER_VALVE_STATUS	Master valve open/closed	0=Closed, 1=Open
00002	WING_VALVE_STATUS	Wing valve open/closed	0=Closed, 1=Open
00003	SWAB_VALVE_STATUS	Swab valve open/closed	0=Closed, 1=Open
00004	CHOKE_VALVE_STATUS	Choke valve open/closed	0=Closed, 1=Open
00005	FLOWLINE_VALVE_STATUS	Flowline valve open/closed	0=Closed, 1=Open
00006	ESD_VALVE_STATUS	ESD valve status	0=Tripped, 1=Normal
00007	ESP_STATUS	ESP run/stop	0=Stopped, 1=Running
00008	WELL_STATUS	Well on/off production	0=Shut-in, 1=Producing

3.6 Alarm Registers (Function Code 02 — Read Discrete Inputs)

Input Address	Tag	Description	Alarm Condition
10001	HIGH_TUBING_PRESSURE_ALARM	Tubing pressure high	1=Alarm
10002	HIGH_HIGH_TUBING_PRESSURE_ALARM	Tubing pressure high-high	1=Alarm
10003	LOW_TUBING_PRESSURE_ALARM	Tubing pressure low	1=Alarm
10004	HIGH_CASING_PRESSURE_ALARM	Casing pressure high	1=Alarm
10005	HIGH_HIGH_CASING_PRESSURE_ALARM	Casing pressure high-high (ESD)	1=Alarm
10006	ESP_OVERLOAD_ALARM	ESP motor overload	1=Alarm
10007	ESP_UNDERLOAD_ALARM	ESP motor underload	1=Alarm
10008	ESP_VIBRATION_ALARM	ESP vibration high	1=Alarm
10009	ESD_ACTIVATED	Emergency shutdown activated	1=Active
10010	COMMUNICATION_FAULT	Field device comm fault	1=Fault

4. DNP3 Object Definitions (KOC-E-027 Section 4.2)

The following DNP3 object definitions are used by the OG RMM Platform Rust edge agent when communicating with KOC legacy SCADA outstations via DNP3 Serial or DNP3/TCP.

4.1 DNP3 Configuration

Parameter	Value
DNP3 Version	IEEE 1815-2012 (DNP3)
Application Layer Timeout	10 seconds
Link Layer Timeout	5 seconds
Unsolicited Reporting	Enabled (Class 1, 2, 3)
Unsolicited Reporting Interval	5 seconds (Class 1), 60 seconds (Class 2/3)
Integrity Poll Interval	300 seconds
Time Sync	Enabled (SNTP)
CRC Error Handling	Retry 3 times, then alarm

4.2 DNP3 Binary Input Objects (Object Group 1)

Index	Tag	Description	Class
0	MASTER_VALVE_STATUS	Master valve open/closed	Class 1
1	WING_VALVE_STATUS	Wing valve open/closed	Class 1
2	ESD_VALVE_STATUS	ESD valve status	Class 1
3	ESP_STATUS	ESP run/stop	Class 1
4	WELL_STATUS	Well on/off production	Class 1
5	HIGH_PRESSURE_ALARM	High pressure alarm	Class 1
6	HIGH_HIGH_PRESSURE_ALARM	High-high pressure alarm (ESD)	Class 1
7	ESP_OVERLOAD_ALARM	ESP overload alarm	Class 1
8	COMMUNICATION_FAULT	Communication fault	Class 1
9	POWER_FAIL_ALARM	Power failure alarm	Class 1

4.3 DNP3 Analog Input Objects (Object Group 30)

Index	Tag	Description	Unit	Class
0	TUBING_PRESSURE	Tubing head pressure	PSI	Class 2
1	CASING_PRESSURE	Casing head pressure	PSI	Class 2
2	FLOWLINE_PRESSURE	Flowline pressure	PSI	Class 2
3	TUBING_TEMP	Tubing temperature	°C	Class 2
4	OIL_FLOW_RATE	Oil production rate	BOPD	Class 2
5	GAS_FLOW_RATE	Gas production rate	MSCFD	Class 2
6	WATER_FLOW_RATE	Water production rate	BWPD	Class 2
7	WATER_CUT	Water cut	%	Class 3
8	ESP_MOTOR_CURRENT	ESP motor current	A	Class 2
9	ESP_MOTOR_TEMP	ESP motor temperature	°C	Class 2
10	ESP_VIBRATION_X	ESP vibration X	G	Class 2
11	ESP_FREQUENCY	ESP drive frequency	Hz	Class 2
12	BATTERY_VOLTAGE	Solar battery voltage	V	Class 3
13	SOLAR_PANEL_CURRENT	Solar panel current	A	Class 3

4.4 DNP3 Binary Output Objects (Object Group 12 — Control Relay Output Block)

Index	Tag	Description	Requires Confirmation
0	MASTER_VALVE_OPEN	Open master valve	Yes (supervisor)
1	MASTER_VALVE_CLOSE	Close master valve	Yes (supervisor)
2	WING_VALVE_OPEN	Open wing valve	Yes (supervisor)
3	WING_VALVE_CLOSE	Close wing valve	Yes (supervisor)
4	ESP_START	Start ESP	Yes (supervisor)
5	ESP_STOP	Stop ESP	Yes (supervisor)
6	ESD_RESET	Reset ESD (after manual inspection)	Yes (supervisor + typed confirmation)

4.5 DNP3 Analog Output Objects (Object Group 41)

Index	Tag	Description	Unit	Range
0	CHOKE_POSITION_SETPOINT	Choke position setpoint	%	0–100
1	ESP_FREQUENCY_SETPOINT	ESP drive frequency setpoint	Hz	30–60
2	WELLHEAD_PRESSURE_SETPOINT	Wellhead pressure setpoint	PSI	100–5,000

5. KOC Tag Naming Convention Mapping

The OG RMM Platform uses a generic `<WELL_ID>-<SENSOR_TYPE>` convention. The following mapping table translates to the KOC-E-027 convention `<FIELD>-<WELL>-<INSTRUMENT>-<SUFFIX>`:

OG RMM Tag	KOC-E-027 Tag	Example
{WELL_ID}-TUBING_PRESSURE	{FIELD}-{WELL}-PT-001-PV	GBK-W047-PT-001-PV
{WELL_ID}-CASING_PRESSURE	{FIELD}-{WELL}-PT-002-PV	GBK-W047-PT-002-PV
{WELL_ID}-OIL_FLOW_RATE	{FIELD}-{WELL}-FT-001-PV	GBK-W047-FT-001-PV
{WELL_ID}-ESP_MOTOR_CURRENT	{FIELD}-{WELL}-IT-001-PV	GBK-W047-IT-001-PV
{WELL_ID}-ESP_VIBRATION_X	{FIELD}-{WELL}-VT-001-PV	GBK-W047-VT-001-PV
{WELL_ID}-MASTER_VALVE_STATUS	{FIELD}-{WELL}-XV-001-ZI	GBK-W047-XV-001-ZI
{WELL_ID}-ESD_ACTIVATED	{FIELD}-{WELL}-XS-001-ZI	GBK-W047-XS-001-ZI

Field codes used by KOC: GBK (Greater Burgan), RWK (Raudhatain/Sabriyah), MNF (Minagish/Umm Gudair), NTH (North Kuwait).

6. KOC Historian Data Retention Configuration

Per KOC-E-027 Section 3.3, the Kuwait Helm profile configures:

Data Type	Retention Period	Storage Class
Raw telemetry (1-second)	2 years	encrypted-ssd-kw
Aggregated (1-minute)	5 years	encrypted-ssd-kw
Aggregated (1-hour)	10 years	standard-kw
Aggregated (1-day)	30 years	archive-kw
Alarm records	10 years	encrypted-ssd-kw
Audit logs	10 years	encrypted-ssd-kw
Financial records	10 years	hsm-encrypted-kw